

Multi View 3D Tree Localization from UAV Imagery Using Segmentation and Ray Triangulation

Diana Emal

School of Computing, University of the Fraser Valley
Abbotsford, BC, Canada
Diana.Emal@student.ufv.ca

Kongwen (Frank) Zhang,

School of Computing, University of the Fraser Valley
Abbotsford, BC, Canada
Frank.Zhang@ufv.ca

Abstract—This paper presents a multi view triangulation pipeline for estimating 3D tree locations from overlapping UAV imagery. Individual trees are segmented using SAM2, and tree centroids are extracted from the resulting masks. Camera poses are estimated using COLMAP, and projection rays from multiple viewpoints are triangulated to estimate 3D tree locations. The method was evaluated on 20 trees observed from multiple viewpoints and produced low ray convergence errors despite the absence of GPS coordinates and real world camera poses. The results demonstrate the feasibility of tree localization using only overlapping UAV imagery.

Keywords—UAV imagery, tree localization, image segmentation, COLMAP, triangulation, 3D reconstruction

I. INTRODUCTION

Urban and environmental tree monitoring has become increasingly important for applications such as urban planning, forestry management, and ecological assessment. Unmanned Aerial Vehicles (UAVs) provide a cost efficient and flexible method for capturing high resolution aerial imagery of the natural environment. However, localizing individual trees in tree dimensional space accurately from overlapping aerial imagery is challenging due to canopy overlap, irregular tree structure, viewpoint variation, and limited geographic information available in many datasets.

Previous studies have explored UAV based deep learning methods for tree detection and localization [1]. However, many approaches rely heavily on GPS information and project detections onto simplified ground representations without explicitly reasoning about multi-view 3D geometry. Other approaches utilize LiDAR or dense Structure-from-Motion (SfM) reconstruction techniques for individual tree analysis [2], but these methods may require specialized sensors, dense reconstruction pipelines, or computationally expensive processing.

This paper proposes a lightweight multi view pipeline for individual tree localization and centroid detection using overlapping high resolution UAV imagery. This method combines image segmentation, centroid extraction, COLMAP based camera pose estimation, and multi view triangulation to estimate stable 3D tree centroid location.

The main contributions of this paper are:

1. A segmentation based multi view triangulation framework for UAV based individual tree localization.
2. Integration of image segmentation and COLMAP camera geometry for estimating consistent 3D locations.

3. Experimental evaluation demonstrating stable triangulation performance across multiple viewpoints.

II. METHODOLOGY

A. Dataset

The dataset used in this research is the `drone_dataset_sheffield_cross` dataset [3]. It consists of 225 overlapping high resolution UAV images captured over an outdoor urban environment containing trees, sports courts, surrounding vegetation, and nearby structures. “Fig. 1” shows one of the UAV image from the dataset. The dataset does not provide GPS information, real world coordinates, or camera pose estimations. Therefore, camera geometry and scene structure were reconstructed using COLMAP. For the experiments conducted in this study, twenty individual trees were manually selected across the dataset, with each tree observed from approximately 7 to 15 different viewpoints.

B. Sparse Reconstruction Using COLMAP

COLMAP is an open source photogrammetry framework that uses Structure from Motion (SfM) techniques to reconstruct 3D scene geometry from overlapping 2D images captured from multiple viewpoints [4]. Since the dataset did not provide camera pose information, COLMAP’s sparse reconstruction pipeline was used to estimate camera positions and generate a sparse 3D point cloud representation of the scene structure [4], as shown in “Fig. 2”. In addition, because the dataset did not contain real world GPS coordinates, tree localization in this study was performed within COLMAP’s arbitrary coordinate system rather than a geo referenced coordinate framework.



Fig. 1. A UAV image of the trees used for this research

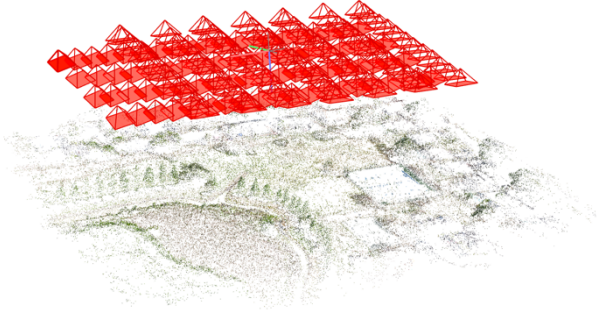


Fig. 2. COLMAP's sparse reconstruction of the scene with camera positions shown in red

C. Tree Segmentation

To isolate individual trees from complex aerial scenes containing multiple trees and surrounding structures, Meta's Segment Anything Model (SAM2) was used to generate binary segmentation masks [5], as shown by "Fig. 3". Individual trees were manually selected using bounding box prompts, after which the centroid of each segmented tree mask was extracted using image spatial moments [6]. The centroid coordinates (x, y) were computed as:

$$x = M_{10} / M_{00} \quad (1)$$

$$y = M_{01} / M_{00} \quad (2)$$

where M_{00} represents the mask area and M_{01} , M_{10} are the first order image moments.

D. Multi View Ray Projection

The extracted tree centroids exist in 2D image space and pixel coordinates. To estimate their corresponding 3D locations, Camera poses recovered from COLMAP [4] were used to construct projection rays following standard multi view geometry formulations [7].

For a 3D point X_{world} in the world coordinate system, COLMAP transforms the point into camera coordinates using a rotation matrix R and translation vector t :

$$X_{\text{camera}} = RX_{\text{world}} + t \quad (1)$$

The camera center C , which represents the origin of the projection ray in world coordinates, is obtained by solving for the world point that maps to the origin of the camera frame:

$$0 = RC + t \quad (2)$$

$$C = -R^{-1} t = -R^T t \quad (3)$$

where $-R^T$ denotes the transpose (inverse) of the rotation matrix.

A ray direction in the camera coordinate frame, $\text{ray}_{\text{camera}}$, was then transformed into the COLMAP world coordinate system using:

$$\text{ray}_{\text{world}} = R^T \text{ray}_{\text{camera}} \quad (4)$$

The resulting world space camera center and ray direction define a 3D projection ray for each tree centroid observation. Each centroid observation therefore produces a unique projection ray originating from its corresponding camera center. The intersection of multiple rays from different viewpoints enables estimation of the tree's 3D location.

E. Triangulation

After constructing camera rays for each tree centroid observation, a least squares triangulation [8] approach was used to estimate the corresponding 3D tree location in COLMAP world coordinates. Each ray is represented by a camera center C_i and a normalized direction vector d_i . Due to segmentation uncertainty, camera pose estimation error, and image noise, the rays do not intersect perfectly in 3D space. Therefore, the goal is to estimate the point X that minimizes the perpendicular distance to all rays simultaneously [8][9].

For a ray direction d_i , the perpendicular projection matrix is defined as:

$$P_i = I - d_i d_i^T \quad (1)$$

where I is the 3x3 identity matrix. The optimal 3D point is obtained by solving the linear least squares system:

$$AX = b \quad (2)$$

where

$$A = \sum P_i \quad (3)$$

$$b = \sum P_i C_i \quad (4)$$

The resulting solution X represents the estimated 3D location of the tree obtained from multiple image observations [9]. "Fig. 4" shows projection ray convergence for one of the trees.

III. EXPERIMENTAL RESULTS

The pipeline was tested on 20 different trees observed from multiple UAV viewpoints. For each tree, multi view triangulation was performed using centroid observations extracted from segmented masks. Mean, median, and maximum ray distances to the estimated 3D point were used to evaluate convergence quality.

"Table I" summarizes the triangulation results for all 20 trees. Across all experiments, the average mean triangulation error was 0.015 COLMAP units, while the highest mean error remained below 0.03 units. These results indicate strong ray convergence and demonstrate that reliable 3D localization can be achieved despite the absence of GPS coordinates and real world camera poses.

TABLE I. SUMMARY OF TRIANGULATION RESULTS FOR 20 TREES

Matric	Value
Trees Evaluated	20
Mean Error (Average)	0.015
Median Error (Average)	0.013
Lowest Mean Error	0.0108
Highest Mean Error	0.0250
Mean Viewpoints Per Tree	10.6



Fig. 3. Segmentation mask generated by SAM2 for a tree

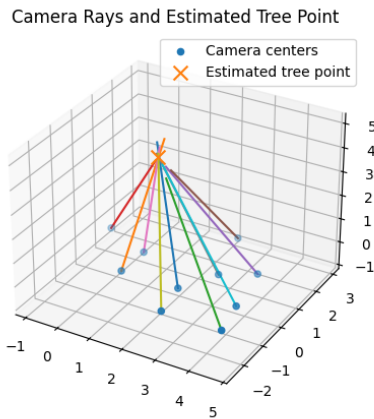


Fig. 4. An example of projection rays converging toward a common 3D point for an individual tree

IV. DISCUSSION

The results obtained from this research demonstrate successful 3D tree localization using UAV imagery captured from multiple viewpoints. Despite the absence of real world GPS coordinates, the proposed pipeline was able to estimate stable 3D tree locations within COLMAP's coordinate system. "Fig. 5" shows that the estimated tree positions were generally accurate in the x and y directions. However, some trees, such as T12, T18, and T4, produced higher z coordinate values than expected. One possible reason is that the actual tree heights and true ground elevation were unknown. In

addition, the localization was performed within COLMAP's arbitrary coordinate system rather than a geo referenced framework, which may introduce inaccuracies in vertical positioning.

One of the main limitations of this research is the dataset itself, which contained a limited number of trees and did not provide real world coordinates or camera pose information. These limitations could be addressed in future improvements of this research.

V. CONCLUSION

This paper presented a multi view triangulation pipeline for estimating 3D tree locations from overlapping UAV imagery. Using SAM2 segmentation, COLMAP based camera pose estimation, and least squares triangulation, stable 3D localization results were achieved across multiple tree observations. Experimental results demonstrated low ray convergence errors despite the absence of GPS coordinates and real world camera poses. The proposed approach shows potential for future applications in UAV based environmental monitoring and 3D tree reconstruction.

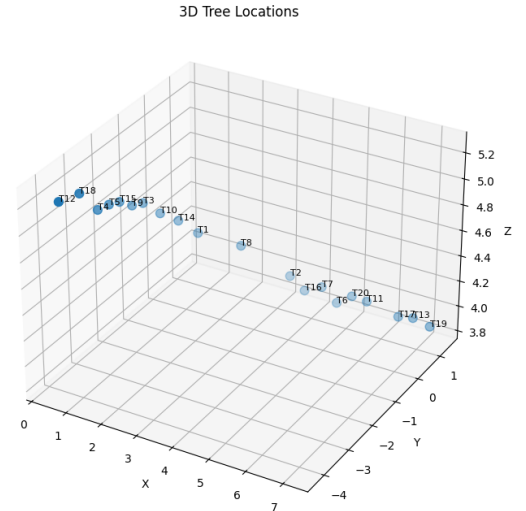


Fig. 5. Estimated tree 3D centroid locations for all 20 trees

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